

Full Stack Development - MERN Stack - syllabus

Duration: 120 hours

Hyper Text Markup Language(HTML5)

- ✓ Introduction to HTML
- ✓ HTML Basics
- ✓ HTML Elements
- ✓ HTML Semantic
- ✓ HTML Attributes
- ✓ HTML Headings
- ✓ HTML Paragraph
- ✓ HTML Styles
- ✓ HTML Formatting
- ✓ HTML Quotations
- ✓ HTML Comments & Colour
- ✓ HTML CSS, Links and Images
- ✓ HTML Lists
- ✓ HTML Blocks
- ✓ HTML Classes
- ✓ HTML Layout
- ✓ HTML Responsive
- ✓ HTML Forms, Tables

Introduction CSS3

- ✓ CSS3 Syntax
- ✓ CSS3 Types
- ✓ CSS3 Color

- ✓ CSS3 Background color
- ✓ CSS3 Padding
- ✓ CSS3 Height
- ✓ CSS3 Width
- ✓ CSS3 Margin
- ✓ CSS3 Shadows
- ✓ CSS3 Align
- ✓ CSS3 Float
- ✓ CSS3 Icon
- ✓ CSS3 DropDown
- ✓ CSS3 Attribute Selector

✓ CSS3 Grid Layout

✓ CSS3 Animation

Introduction to JavaScript

- ✓ History of JavaScript
- ✓ JavaScript Code Execution
- ✓ Console in JavaScript
- ✓ JavaScript Versions
- ✓ JavaScript Syntax
- ✓ JavaScript Comments
- ✓ Where to Put JavaScript in HTML
- ✓ JavaScript First Program Hello world
- ✓ JavaScript data types
- ✓ **JavaScript variables**

- ✓ let Keyword
- ✓ var Keyword
- ✓ const Keyword
- ✓ **JS Expression and Operators**
- ✓ JS Assignment operators
- ✓ JS Comparison operators
- ✓ JS Arithmetic operators
- ✓ JS Bitwise operators
- ✓ JS Logical operators
- ✓ JS BigInt Operators
- ✓ JS String operators
- ✓ JS Ternary operator
- ✓ JS Relational operators
- ✓ **JavaScript Control Flow Statements**
- ✓ JS if Statement
- ✓ JS if...else Statement
- ✓ JS if else if Statement
- ✓ JS Nested if Statement
- ✓ JS switch Statement
- ✓ JS try...catch Statement
- ✓ **JavaScript Loops**
- ✓ JS for Loop
- ✓ JS do...while Loop
- ✓ JS while Loop

- ✓ JS for...in Loop
- ✓ JS for...of Loop
- ✓ JS labeled Statement
- ✓ JS break Statement
- ✓ JS continue Statement
- ✓ **JavaScript Objects**
- ✓ JS Array
- ✓ JS String
- ✓ JS Date
- ✓ JS Number
- ✓ JS Math
- ✓ JS Object
- ✓ JS Boolean
- ✓ JS JSON
- ✓ JS Map
- ✓ JS Get,Set
- ✓ JS WeakMap
- ✓ JS WeakSet
- ✓ **JavaScript Functions**
- ✓ JS Functions in JavaScript
- ✓ JS Function Definitions
- ✓ JS Function Call
- ✓ JS Function Expression
- ✓ JS Function Parameters

- ✓ JS Anonymous Functions
- ✓ JS Arrow functions
- ✓ JS Nested functions
- ✓ JS Async/Await Function
- ✓ **JavaScript Regular Expression**
- ✓ JS Regular expressions
- ✓ JS RegExp [abc] Expression
- ✓ JS RegExp S Metacharacter
- ✓ JS RegExp m Modifier
- ✓ JS RegExp ?! Quantifier
- ✓ JS RegExp {X,Y} Quantifier
- ✓ JS RegExp test() Method
- ✓ JS RegExp [^0-9] Expression
- ✓ **JavaScript Events**
- ✓ JS Events
- ✓ JS onclick Event
- ✓ JS dblclick Event
- ✓ JS onload Event
- ✓ JS onchange Event
- ✓ JS Focus Event
- ✓ JS onscroll Event
- ✓ **JavaScript OOPs**
- ✓ JS Classes In JavaScript
- ✓ JS JavaScript class expression

- ✓ JS Object Constructors
- ✓ JS Static Methods
- ✓ JS JavaScript Constructor Method
- ✓ JS JavaScript Encapsulation
- ✓ JS Inheritance
- ✓ JS Polymorphism
- ✓ JS Abstraction
- ✓ JS Inheritance and Prototype Chain
- ✓ Prototype Inheritance in JavaScript
- ✓ JavaScript Memory Management
- ✓ JavaScript Iterators and generators
- ✓ JS Iterator
- ✓ JS Function Generator
- ✓ JavaScript Validations
- ✓ JS Form Validation
- ✓ JS Email Validation
- ✓ JavaScript Exception Handling
- ✓ JS Exception Handling
- ✓ JS try-catch Statement
- ✓ JS throw

Introduction to Bootstrap

- ✓ Bootstrap Basics
- ✓ Bootstrap CSS, Bootstrap JS

jQuery, JSON

ReactJS

- ✓ React Introduction
- ✓ React Environment Setup
- ✓ **ReactJS Components**
- ✓ ReactJS Components
- ✓ ReactJS Container and Presentational Pattern in Components
- ✓ ReactJS Functional Components
- ✓ React Class Components
- ✓ Differences between Functional Components and Class Components
- ✓ **React JSX**
- ✓ ReactJS Props and States
- ✓ ReactJS Methods as Props
- ✓ ReactJS PropTypes
- ✓ ReactJS Props - Set 1
- ✓ ReactJS Props - Set 2
- ✓ ReactJS Unidirectional Data Flow
- ✓ ReactJS State
- ✓ **ReactJS State vs Props**
- ✓ ReactJS Babel Introduction
- ✓ ReactJS Key Concepts
- ✓ ReactJS Virtual DOM
- ✓ React JS ReactDOM
- ✓ ReactJS Lists

✓ React Forms

- ✓ ReactJS Handling forms
- ✓ ReactJS Submitting forms
- ✓ ReactJS Multiple input fields

✓ ReactJS Lifecycle of Components

✓ ReactJS Hooks

- ✓ React Hooks
- ✓ React useState Hook
- ✓ ReactJS useEffect Hook
- ✓ React JS useRef Hook
- ✓ ReactJS useContext Hook

✓ React Custom Hooks

✓ ReactJS Advance Concepts

- ✓ React Router
- ✓ React JS Types of Routers
- ✓ How to fetch data from an API in ReactJS ?
- ✓ Explain new Context API in React

✓ ReactJS Hands-On Tasks

- ✓ ReactJS | Calculator App (Introduction)
- ✓ ReactJS Calculator App (Structure)
- ✓ ReactJS Calculator App (Building UI)
- ✓ ReactJS Calculator App (Adding Functionality)
- ✓ ReactJS Calculator App (Styling)
- ✓ ReactJS Connection and Deployment

Node.js

- ✓ Node.js Introduction
- ✓ Node.js --save and --save-dev
- ✓ **Data Type**
- ✓ Node.js Data Type
- ✓ Node.js Numbers
- ✓ Node.js Booleans
- ✓ Node.js Strings
- ✓ Node.js String Functions
- ✓ Node.js Callback Concept
- ✓ Node.js server
- ✓ **Buildin Object**
- ✓ Node.js Objects
- ✓ Node.js JSON
- ✓ Node.js Arrays
- ✓ Node.js Array Functions
- ✓ **Function**
- ✓ Node.js Functions
- ✓ Node.js Higher Order Function
- ✓ **Language**
- ✓ Node.js Operator
- ✓ Node.js Loop Statement
- ✓ **Object**
- ✓ Node.js Class

- ✓ Node.js Class Creation
- ✓ Node.js prototype chain
- ✓ Node.js Errors and Exceptions
- ✓ Node.js Globals
- ✓ **Advanced**
- ✓ Node.js Asynchronous Programming
- ✓ Node.js Web
- ✓ Node.js Buffer
- ✓ Node.js process
- ✓ Node.js setTimeout setInterval
- ✓ Body parser
- ✓ **Module**
- ✓ Node.js Module System
- ✓ Node.js Module Require
- ✓ Node.js File Modules
- ✓ Node.js Path Module
- ✓ Node.js bcrypt Module
- ✓ Node.js os Module
- ✓ Node.js util Module
- ✓ **Package**
- ✓ Node.js Packages
- ✓ Node.js package.json File
- ✓ **NPM**
- ✓ Node.js NPM JSON

- ✓ Node.js NPM
- ✓ **Event**
- ✓ Node.js Events
- ✓ Node.js Event loop
- ✓ Node.js Event Emitter
- ✓ **Stream**
- ✓ Node.js Streams
- ✓ Node.js Readable/Writable Streams
- ✓ Node.js Custom Streams
- ✓ **HTTP**
- ✓ Node.js HTTP
- ✓ Node.js HTTP Headers
- ✓ Node.js HTTP Files

Mongo DB

- ✓ Introduction to MongoDB
- ✓ MongoDB Environment
- ✓ MongoDB Create Database
- ✓ MongoDB Drop Database
- ✓ MongoDB Create Collections
- ✓ MongoDB Drop Collections
- ✓ MongoDB Read Operation
- ✓ MongoDB Write Operations
- ✓ MongoDB Data Modeling

- ✓ MongoDB Administration
- ✓ MongoDB Atlas - Cloud configuration
- ✓ MongoDB Security
- ✓ MongoDB Schema
- ✓ MongoDB Indexes
- ✓ MongoDB Storage
- ✓ MongoDB Replications

Express js

- ✓ Introduction to Express Framework
- ✓ Getting Started with Express Your first Express App
- ✓ Express Routing - Implementing MVC in Express, Middleware- Using Template - Engines
- ✓ Error Handling - API Handling
- ✓ Developing Template Engine - Using Process Managers
- ✓ Security & Deployment
- ✓ SEO, DNS, HOST, SMTP Basics
- ✓ Payment Gateway Project
- ✓ Auth Project
- ✓ Real- Time Projects on MERN Stack(Rest - API)

Guidelines For A Successful Project